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Paleomagnetism and Mesozoic Tectonics of the Seven Devils Volcanic Arc in Northeastern Oregon

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Upper Triassic volcanogenic rocks of the Seven Devils Group and Huntington Formation of Brooks (1979b) in northeastern Oregon record two episodes of magnetization. The older magnetization, which predates Late Jurassic deformation of the volcanic arc terrane comprising the Seven Devils Group, was probably acquired in Late Triassic time. The inclinations of both normal and reversed polarity magnetizations indicate that the Triassic rocks originated approximately 18° from the paleoequator. The results are consistent with paleomagnetic data from similar Triassic rock assemblages on Vancouver Island and in south central Alaska. We interpret these assemblages as displaced fragments of Wrangellia, a once coherent late Paleozoic volcanic arc and early Mesozoic volcanic plateau that was torn apart and carried north by plate motion along the continental margin. We interpret the younger magnetization, which fails the fold test, as a widespread thermochemical overprint caused by Late Jurassic and Early Cretaceous plutonism. This magnetization yields a paleomagnetic pole at 60°N, 295°E (α_{95} = 7°), and is similar to the pole obtained from plutons that intrude the Seven Devils Group. The angular difference between the pole from the plutons and the Late Jurassic–Early Cretaceous reference pole for stable North America suggests that the Seven Devils volcanic arc has undergone a large clockwise rotation (66° ± 21°). The rotation may reflect the last stages of the arc's accretion to the continental margin or later large-scale block movements in the Pacific Northwest.

INTRODUCTION

Pre-Cenozoic rocks in eastern Oregon, eastern Washington, and western Idaho can be placed in two distinct geologic settings approximately separated by the strontium-isotope line (Figure 1) of *Armstrong et al.* [1977]. East of the line ($^{87}Sr/^{86}Sr > 0.706$) is the miogeocline, consisting of Paleozoic sedimentary rocks that were deposited on Precambrian rocks of the North American craton. These rocks were intruded by the Cretaceous and Eocene Idaho batholith. West of the line ($^{87}Sr/^{86}Sr < 0.706$) is a complex terrane, apparently made up of oceanic crust and fragments of volcanic arcs, that ranges from Devonian to Early Cretaceous in age. Plutons intruded and essentially welded the fragments together, then the terrane was overlapped by Upper Cretaceous marine strata.

The western terrane, part of the Cordilleran 'collage' of *Davis et al.* [1978], is composed of four smaller terranes (Figure 2) that were apparently brought together by plate movements [*Brooks et al.*, 1976; *Vallier et al.*, 1977; *Brooks and Vallier*, 1978; *Dickinson and Thayer*, 1978; *Brooks*, 1979*a*]. These terranes are (1) the northern Wallowa-Seven Devils volcanic arc terrane, consisting of Lower Permian and Middle and Upper Triassic volcanogenic rocks overlain by Upper Triassic to Upper Jurassic marine sedimentary rocks, (2) the central oceanic crust terrane [*Brooks and Vallier*, 1978] or melange [*Dickinson and Thayer*, 1978] of dismembered ophiolite, argillite, tuff, and limestone that ranges in age from Devonian to Late Triassic, (3) Upper Triassic to Upper Jurassic marine sedimentary rocks that are typical of a forearc basin deposit [*Dickinson and Thayer*, 1978], and (4) the Upper Triassic Huntington Formation of *Brooks* [1979*b*], a volcanic arc terrane [*Brooks et al.*, 1976; *Brooks and Vallier*, 1978] that may be either a separate fragment or the southern extension of the Wallowa-Seven Devils volcanic arc. The Wallowa-Seven Devils volcanic arc and the Huntington volcanic arc terranes are shortened in this paper to the 'Seven Devils arc' and the 'Huntington arc,' respectively.

These four terranes are knitted together by a large number of plutons of Late Jurassic and Early Cretaceous age, thereby indicating that the complex amalgamation occurred before the Late Jurassic. The Late Cretaceous and early Tertiary was a period of uplift, volcanism, and erosion, followed in the Miocene by a tremendous outpouring of flood basalts, the Columbia River Basalt Group [e.g., *Walker*, 1979].

Little is known about the origins of the amalgamated terranes. One hypothesis which can be tested by paleomagnetic data is that the terranes formed far from their present positions relative to North America. After being displaced great distances by plate motions, the terranes in question were accreted to the continental margin. Any large-scale shift in latitude undergone by the exotic terranes could be detected by paleomagnetic methods.

The object of our study was to investigate possible displacement between the Seven Devils and Huntington arcs and the craton, by comparing paleomagnetic data from Middle and Upper Triassic rocks in the arcs with coeval data from central North America. We also wanted to compare the Triassic paleolatitude of the Seven Devils and Huntington arcs with results from volcanic rocks of identical age in Alaska and British Columbia, to test whether all originated at a common latitude. Specifically, the paleomagnetic comparison is made between Triassic rocks of the Seven Devils arc with those of Vancouver Island (British Columbia), the Wrangell Mountains (south-central Alaska), and the central Alaska Range, where nearly identical lithologies and depositional histories occur. These stratigraphic packages are part of 'Wrangellia,' a geologic terrane characterized by Pennsylvanian and Lower Permian volcanogenic rocks overlain by Middle and (or) Upper Triassic basaltic rocks up to 3 km thick [*Jones et al.*, 1977]. The volcanic rocks are overlain by platform carbonates of Karnian and Norian ages, which in turn are overlain by Upper Triassic and Lower Jurassic

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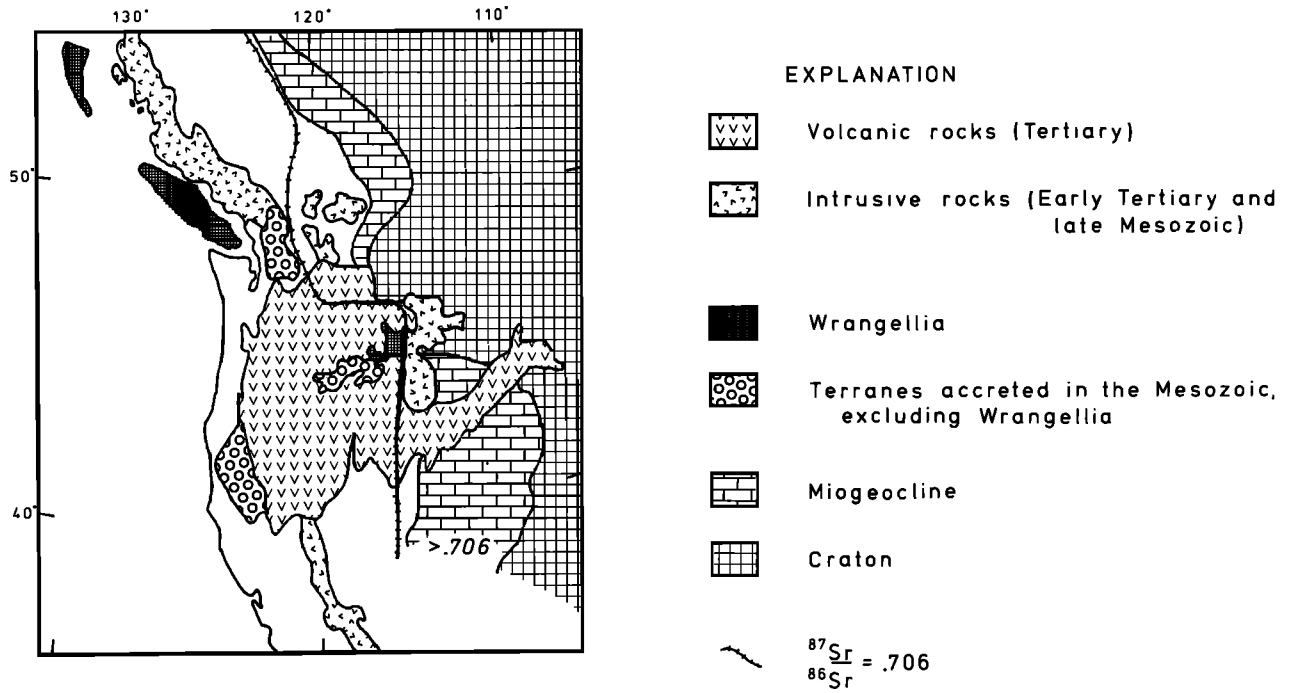


Fig. 1. Geologic provinces of northwestern North America.

marine sedimentary rocks. Evidence derived from stratigraphy, paleontology, and paleomagnetism suggests that Wrangellia consists of displaced fragments of a formerly coherent volcanic and sedimentary terrane.

The exotic origin of northern Wrangellia is well documented by paleomagnetic results from the Nikolai Greenstone [*Hillhouse*, 1977] and equivalent basalt in the Healy quadrangle [*Hillhouse and Gromme*, 1981], all in south-central Alaska. The discrepancy between predicted and observed magnetic latitudes indicates that since the Late Triassic the Alaskan fragments of Wrangellia have moved northward thousands of kilometers relative to central North America. A similar anomalously low paleolatitude was determined from another part of Wrangellia, the Karmutsen Formation, which is a thick sequence of Upper Triassic basalt on Vancouver Island [*Schwarz et al.*, 1980; *Yole and Irving*, 1980]. The Seven Devils volcanic arc, which has Wrangellia stratigraphy, was interpreted as a displaced companion of the Alaskan and British Columbian fragments [*Jones et al.*, 1978]. Throughout Wrangellia, from the Seven Devils Group to Alaska, the transition from basalt to nonvolcanic calcareous rocks occurs at about the same age, that is, in the late Karnian. The Huntington Formation evidently overlaps in age with the upper part of the Seven Devils

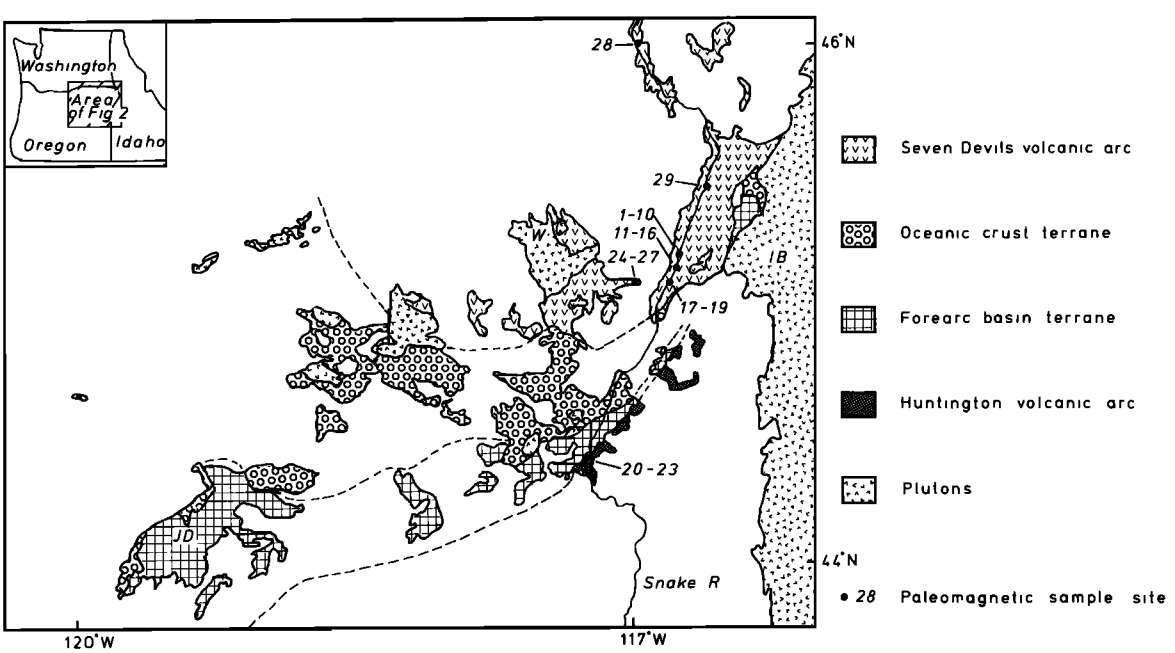


Fig. 2. Geologic terranes of the Blue Mountains province, Oregon and Idaho [after *Brooks*, 1979*a*]. Paleomagnetic sample localities (dots) are keyed to Tables 1 and 2; W, Wallowa batholith; IB, Idaho batholith; JD, John Day inlier.

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Group, but it includes volcaniclastic rocks as young as late middle Norian. It has a significantly different volcanic and depositional history as compared with any part of Wrangellia (N. J. Silberling, written communication, 1981).

Our new paleomagnetic results indicate that the Seven Devils Group and the Huntington Formation share a common paleolatitude with the displaced fragments of Wrangellia. An unexpected result of our study was the discovery of a regional magnetic overprint, presumably related to the intrusion of plutons during the Late Jurassic and Early Cretaceous. The paleomagnetic pole corresponding to the overprint supports the conclusion of *Wilson and Cox* [1980] that the Blue Mountains province has rotated clockwise since 130 m.y. ago. Their results, obtained from the 150–130 m.y. old plutons, indicate a rotation of 66° ± 21° relative to the craton.

STRATIGRAPHY

Our paleomagnetic sampling is concentrated in the Middle and Upper Triassic volcanic flow and sedimentary rocks of the Seven Devils and Huntington arcs. Most samples were selected from the steep walls of the spectacular Snake River Canyon, where the river has cut through the Columbia River basalt group and deep into the underlying pre-Cenozoic rocks. Outcrops along the Imnaha River Canyon also were sampled.

In the Seven Devils arc, several thousand meters of Permian, Triassic, and Jurassic rocks are exposed [*Vallier,* 1974, 1977]. Four major rock stratigraphic units are recognized: (1) the Lower Permian and Middle and Upper Triassic Seven Devils Group, (2) the Upper Triassic (mostly Norian) Martin Bridge limestone, (3) the Upper Triassic and Lower Jurassic Hurwal Formation, and (4) the Upper Jurassic Coon Hollow Formation. The Seven Devils Group has four formations, including the Permian(?) Windy Ridge Formation, the Lower Permian Hunsaker Creek Formation, the Middle and Upper Triassic (Ladinian and Karnian) Wild Sheep Creek Formation, and the Upper Triassic (Karnian and Norian) Doyle Creek Formation. The Permian and Triassic rocks are separated along an unconformity that represents about 20–40 m.y. Most of our paleomagnetic samples in the Seven Devils arc were selected from the Wild Sheep Creek Formation.

The Wild Sheep Creek Formation [*Vallier,* 1977] is a thick unit of metamorphosed basalt, basaltic andesite, andesite, dacite flows and volcaniclastic rocks, graywacke, argillite, and limestone that is widely exposed in the Snake River Canyon, the Seven Devils Mountains, and the Wallowa Mountains. The formation is characterized by strata that thicken and thin along strike or pinch out over short distances, by rapid changes in lithofacies, and by extensive dikes and sills.

In the Huntington arc the Upper Triassic Huntington Formation [*Brooks,* 1979*b*] was sampled for paleomagnetic studies. According to *Brooks et al.* [1976], this rock unit comprises lavas and volcaniclastic rocks, coarse- to fine-grained sedimentary rocks, minor limestone, and small intrusive bodies. As mentioned above, the unit is of Karnian and Norian age and may be equivalent to parts of the Wild Sheep Creek and Doyle Creek formations of the Seven Devils arc. The overlying Weatherby Formation of *Brooks* [1979*b*] is mostly wacke, siltstone, and argillite with minor phyllite, slate, conglomerate, limestone, and gypsum. The age of the unit is Early and Middle Jurassic (Sinemurian to Callovian).

Paleomagnetic Sampling and Lithology

We collected oriented cores at 25 sites in the Wild Sheep Creek Formation and at four sites in the Huntington Formation (Figure 2). A site is defined as either a single flow represented by 3 to 10 cores or a stratigraphic section (2–100 m thick) represented by 5 to 39 cores. All but a few sites were taken in fresh road cuts. The Wild Sheep Creek and Huntington formations are in places extensively faulted and deformed; therefore, sites were located only in strata with well-defined bedding.

Lithologies, approximate flow or stratigraphic unit thicknesses, and structural attitudes of the sites are summarized in Table 1. In general, the volcanic flow rock samples were taken in dark-green and greenish-black submarine basalt, andesite, and dacite flows from 5 to 10 m thick. Most flow rocks have porphyritic textures, and all have been metamorphosed to the greenschist facies. Dominant primary minerals are plagioclase, augite, quartz, and iron ore minerals. Secondary minerals are albite, quartz, chlorite, epidote, calcite, white mica, and hematite. Groundmasses, most of which were originally glassy or tachylitic, are replaced by secondary minerals, particularly chlorite.

The sedimentary rocks that were sampled include a wide variety of volcaniclastics, such as tuff, breccia, and sandstone. Most of the sampled sedimentary rocks have red or purple weathered surfaces, and freshest material usually has a red matrix. Microscopic examination of polished sections reveals abundant authigenic hematite as reddish pigment or as microcrystals that replace primary minerals and coat framework grains. The pyroclastic rocks were originally composed of volcanic rock fragments, primary minerals, glass shards, and pumice that were derived from nearby volcanoes and deposited in a shallow marine environment. In addition to the pyroclastic rocks there are abundant epiclastic beds of conglomerate, breccia, graywacke, and siltstone. These rocks contain volcanic rock fragments, feldspar, and quartz in a matrix of calcite and chlorite; some beds are arkosic.

Vallier and Batiza [1978] showed that there had been extensive major element mobility in the volcanic flow rocks, although, in general, there had been a mass balance of most components. Na_2O, CO_2, and H_2O probably were added to many of the rocks, but the sources of those components were not firmly established. CaO, MgO, FeO, and SiO_2 also may not have been conserved. The rocks originally were basalt, basaltic andesite, andesite, dacite, and rhyolite, most relatively low in K_2O, that are very similar to present-day intraoceanic island arcs that have both tholeiitic basalts and rocks that range in composition from basalt to rhyolite along a calc-alkaline differentiation trend.

Analysis of Remanent Magnetism

Natural remanent magnetizations (NRM) of cylindrical specimens (about 11 cm^3) were measured with a three-component cryogenic magnetometer. The stability of remanence and presence of multicomponent magnetizations were investigated by progressive demagnetization treatments that employed thermal and alternating field methods. Demagnetization in alternating fields up to 100 mT was made with an unshielded three-axis tumbler or with a shielded, fixed-

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TABLE 1. Descriptions and Structural Attitudes of Sample Sites

Site	Latitude, N deg	Longitude, E deg	Description	Structural Attitude: Strike, deg	Structural Attitude: Dip, deg
			Wild Sheep Creek Formation, Hells Canyon, Seven Devils Arc		
1	45.24	243.30	metabasalt flow	220	22 NW
2	45.24	243.30	metabasalt flow	220	22 NW
3	45.24	243.30	metabasalt flow	225	41 NW
4	45.24	243.30	volcanic breccia; red, massive, 6 m thick	225	41 NW
5	45.24	243.30	metabasalt flow	225	41 NW
6	45.24	243.30	metabasalt flow	225	41 NW
7	45.24	243.30	tuffaceous breccia; green, 5 m thick	242	45.5 NW
8	45.24	243.30	volcaniclastic sandstone; red, 2 m thick	242	45.5 NW
9	45.23	243.30	metabasalt flow	235	35 NW
10	45.23	243.30	metabasalt flow	235	35 NW
11	45.20	243.29	metadacite	200	26 W
12	45.19	243.29	meta-andesite dike	200	26 W
13	45.17	243.28	volcaniclastic sandstone; red, 50 m thick	233	60 NW
14	45.17	243.28	volcaniclastic sandstone; red, thinly bedded, 15 m thick	255	42 N
15	45.16	243.28	volcaniclastic sandstone; red, 20 m thick	200	52 W
16	45.16	243.28	tuffaceous breccia; red, well bedded, 15 m thick	320	25 NE
17	45.12	243.25	volcaniclastic sandstone and breccia; red, well-bedded, 5 m thick	339	65 NE
18	45.11	243.25	volcanic breccia; red, massive, 3 m thick	306	68 NE
19	45.11	243.23	volcaniclastic sandstone; red, 5 m thick	020	14 E
			Huntington Formation of Brooks [1979b], Huntington Arc		
20	44.45	242.78	volcaniclastic sandstone; orange, siltstone interbeds, 5 m thick	253	68 N
21	44.45	242.78	metabasalt flow	253	68 N
22	44.45	242.78	volcaniclastic sandstone; red, massive, 3 m thick; and metabasalt flow	253	68 N
23	44.45	242.78	volcanic breccia; red-purple, well bedded, 10 m thick	258	48 N
			Wild Sheep Creek Formation, Imnaha River, Seven Devils Arc		
24	45.11	243.01	basalt dike (Tertiary?)	010	14 E
25	45.11	243.05	volcanic breccia; red, well-bedded, 2 m thick	320	48 NE
26	45.11	243.06	metabasalt flow	335	25 NE
27	45.11	243.06	metabasalt flow	335	25 NE
			Wild Sheep Creek Formation, Lower Hells Canyon, Seven Devils Arc		
28	46.06	243.05	volcaniclastic siltstone; red, interbedded with sandstone, 5 m thick	150	30 W
29	45.57	243.45	volcanic breccia and metabasalt; gray-green, 100 m thick	324	27 NE

specimen apparatus. The median destructive field (MDF) is defined as the peak alternating field required to reduce the NRM to one half its initial magnitude. Thermal demagnetizations up to 680°C were made in air in a magnetically shielded furnace that had an internal field of less than 10 nT. Throughout this paper, blocking temperature (T_B) is defined as the temperature that demagnetizes the NRM of a specimen after being heated approximately 15 min in the field-free furnace. In a typical specimen the NRM will decay over a wide range of temperatures that define the T_B distribution. Our usage of T_B applies to thermal demagnetization in the laboratory and does not imply that the NRM was acquired at that temperature in nature.

Approximately 25% of the specimens were given stepwise demagnetization treatments employing 10 to 14 steps. At many sites, thermal and alternating field methods were applied to companion specimens cut from the same cores so comparisons could be made between distributions of blocking temperatures and median destructive fields. Vector diagrams, that depict the progressive decay of the components of magnetization, established whether stable end points had been reached during the demagnetization runs. From the vector diagrams we found that many specimens exhibited two superposed magnetizations which could be isolated by calculating vector differences between successive demagnetization steps.

From the wide variety of rock types sampled in the Huntington and Wild Sheep Creek formations we identified

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TABLE 2. Site-Mean Paleomagnetic Results From Triassic Rocks of the Wild Sheep Creek Formation and Huntington Formation of *Brooks* [1979*b*]

Site	Treatment	I	D	I_c	D_c	R	k	α_{95}	N	(N_c)
					Group 1					
7	619°–653°	−37.3	27.6	−48.2	72.1	4.9761	168	5.9	5	(6)
8	660°–664°	−56.5	61.7	−35.9	109.1	3.9948	576	3.8	4	(4)
9	661°–664°	−40.2	62.5	−28.1	85.8	5.9535	108	6.5	6	(6)
10	643°–664°	−29.8	68.8	−16.8	83.4	4.9908	434	3.7	5	(5)
18	626°–652°	36.8	71.5	−22.4	66.1	5.8846	43	10.3	6	(6)
					Group 2					
13a	619°–670	65.6	194.5	42.7	296.8	14.5337	30	7.1	15	(18)
20	499°–624°	77.9	182.3	33.3	338.2	5.8960	48	9.8	6	(7)
22	565°–602° 100 mT	65.1	231.3	29.0	316.4	8.7811	37	8.6	9	(9)
23	60 mT	59.9	270.4	30.0	313.6	10.5714	23	9.6	11	(11)
25	30 mT	29.0	311.6	24.8	337.6	7.6885	22	11.9	8	(8)
26	40 mT	22.4	292.5	37.6	304.3	5.9674	153	5.4	6	(10)
					Group 3					
1	20 mT	65.4	356.7	47.4	336.6	7.9931	1009	1.7	8	(8)
2	30 mT	78.6	8.6	60.5	330.0	7.9270	96	5.7	8	(8)
3	15 mT	67.9	26.7	38.4	342.1	6.9700	200	4.3	7	(8)
4	607°–631°	74.2	18.4	40.3	333.6	7.9575	165	4.3	8	(8)
5	20 mT	68.1	16.3	35.7	338.7	6.9297	85	6.6	7	(8)
6	10 mT	71.2	65.1	51.9	344.3	7.9761	293	3.2	8	(8)
11	20 mT	71.8	47.8	66.6	334.0	7.8088	37	9.3	8	(8)
12	30 mT	76.1	31.9	63.4	321.7	2.9834	120	11.3	3	(3)
13b	620°–662°	66.0	57.8	29.1	350.6	23.2961	33	5.3	24	(24)
14	599°–662°	60.2	46.3	29.0	14.9	21.5763	15	8.0	23	(27)
15	626°–642°	81.4	63.0	43.6	298.7	4.9351	62	9.8	5	(5)
16	615°–642°	81.3	90.6	57.9	60.7	22.8305	130	2.7	23	(23)
17	552°–629°	52.5	35.3	−7.1	49.1	6.8547	41	9.5	7	(8)
19	599°–615°	66.9	58.1	56.5	76.0	8.9408	135	4.4	9	(9)
27	30 mT	71.9	27.5	49.2	48.2	6.9455	110	5.8	7	(8)
28	600°–642°	67.6	34.1	76.4	285.1	10.8029	51	6.5	11	(11)
29	30 mT	76.7	74.7	50.3	61.3	15.3637	24	7.8	16	(16)

Treatment, thermal cleaning given in degrees C; AF cleaning in milli Tesla; I, inclination (positive down); D, declination east of north; I_c, D_c, inclination and declination corrected for structural attitude; R, resultant of N unit vectors; k, α_{95}, Fisher precision parameter and 95% confidence radius; N (N_c), number of specimens averaged (collected). Sites 21 and 24 omitted from calculations, see text.

the following families of magnetic directions; each probably related to a thermal event in the geologic history of the region: Group 1, characterized by high blocking temperatures (T_B) and high median destructive fields (MDF), consists of easterly directions with low, negative inclinations when corrected for tilt of the bedding. Group 2, also characterized by high T_B and MDF, lies in the northwest quadrant at low positive inclinations. Group 3, which is present in the majority of specimens, spans a broad range of T_B and MDF and was acquired after the rocks were deformed as indicated by a fold test. Group 3 yields well-grouped steep inclinations in the northeast quadrant before structural correction. The isolation of these components, their T_B and MDF ranges, and their geographic distributions are described in the following sections.

Group 1

Thermal demagnetization of specimens from five sites (7, 8, 9, 10, 18) produced stable end points that define this group of magnetic directions (Table 2, Figure 3). The sites are located in road cuts on the Idaho Power and Light Company (IPALCO) road in Hells Canyon; one group is near Steamboat Creek and the other is 10 km south, near Hibble Gulch. Lithologies consist of green tuffaceous breccia, red volcaniclastic sandstone, red volcanic breccia, and metabasalt flows.

Two examples from the pilot thermal demagnetization runs typify the thermal properties of NRM measured at these five sites. In these examples we denote the NRM and demagnetized vectors obtained from progressive heating steps as J_{20}, J_{100}, etc. Example 1 (Figure 4), a red sandstone, has essentially a single-component magnetization, as is demonstrated on the vector diagram by the linear decay of magnetization to the diagram's origin. The magnetic direction remains stable during heating up to 674°C, the temperature at which 30% of the initial magnetization has been removed. Approximately 95% of the remanence resides in magnetic domains that have T_B exceeding 585°C, the Curie temperature of magnetite. In contrast, example 2, a metabasaltic tuff, exhibits two components of magnetization (Figure 5). Both projections on the vector diagram can be approximated by two line segments, each corresponding to a direction of magnetization: one segment (J_{20}–J_{545}) diverges from and the other (J_{545}–J_{674}) converges to the origin of the diagram, indicating a stable end point is reached at temperatures greater than 545°C. The sharp intersection of the segments at 545°C indicates that the T_B distributions of the two components of magnetization do not overlap appreciably. The T_B distribution of the low-temperature component is calculated by subtracting the vector J_{545} from the vectors obtained at lower temperatures. Heating the specimen to 545°C erases the low-T_B component, so the distribution of the remaining component is calculated from the decay of the vector J_{545}. Most of the magnetization of the high-tempera-

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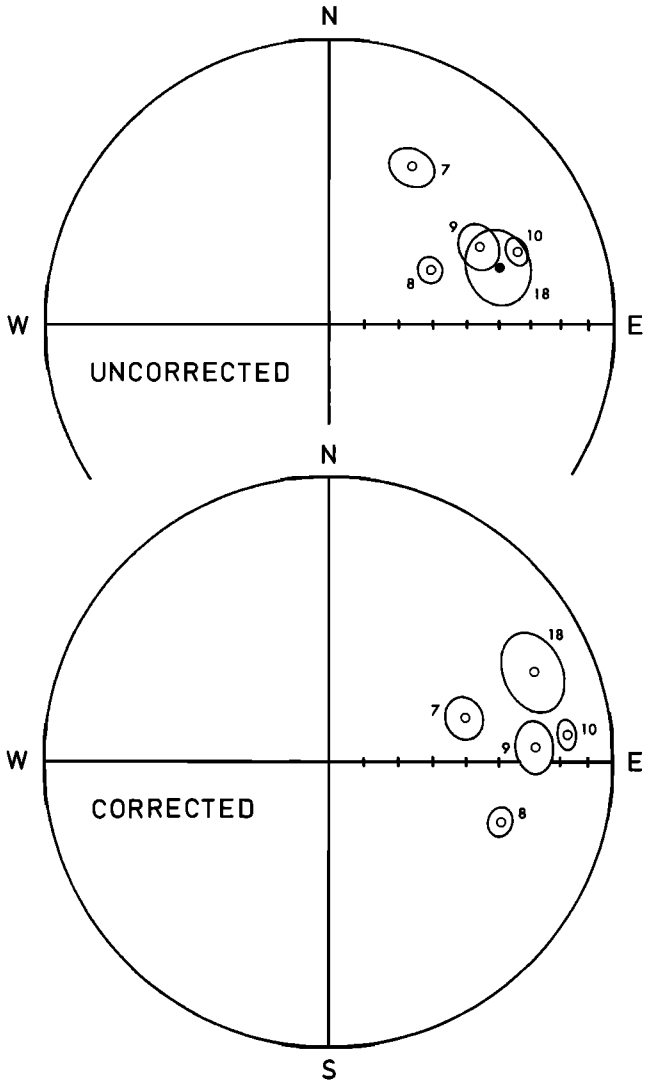


Fig. 3. Mean magnetic field directions from sites in group 1, before and after application of tilt corrections; open circles, upward inclinations; solid circles, downward inclinations. All sites in Hells Canyon.

ture component resides in domains with T_B exceeding 635°C, while the distribution of the low-T_B component is mostly below 300°C.

Thermal cleaning of all specimens included in this group of sites revealed T_B distributions that ranged from the single-component magnetization of example 1 to the two-component system of example 2. The group 1 direction corresponds to the high-T_B magnetization isolated by heating all specimens above 600°C. Vector diagrams depicting the decay of remanence during thermal cleaning showed, in all cases, that the magnetizations reached stable end points. The low-T_B magnetization, which varies widely in intensity from specimen to specimen, is mostly part of group 3, discussed in a following section.

A number of companion cores were AF demagnetized for comparison with the thermal demagnetization results. We found that the vector removed by AF's up to 50 mT roughly corresponded to the vector removed by heating to 545°C and that the high T_B magnetization was unaffected by AF treatment to 60 mT or more.

After correction for tilt of the bedding, the site means of group 1 are clustered to the east and have inclinations near −30° (Figure 3). The apparent decrease in directional scatter produced by the bedding corrections suggests a positive fold test; however, the Fisher precision parameters of the two populations are not significantly different at the 95% confidence level according to the F test [*Irving*, 1964]. The small number of independent structural corrections (N = 3) precludes a significant test. Nevertheless, the bedding correction reduces the spread in inclination from 93° to 31°, a considerable improvement that suggests group 1 directions predate deformation of the region.

Group 2

Magnetizations in this category were detected in the Huntington Formation and in the Wild Sheep Formation (Table 2, Figure 6). At Brownlee Reservoir we sampled a section of the Huntington Formation, which is composed of red volcaniclastic sandstone, volcanic breccia, and metabasalt flows (sites 20, 21, 22, 23). Red volcanic breccia and metabasalt exposed along the Imnaha River (sites 25 and 26) and red sandstone near Squaw Creek (site 13a) are parts of the Wild Sheep Creek Formation that yielded group 2 directions.

In general, NRM's of the Huntington samples consist of two or more components having T_B distributions that overlap between 200° and 400°C, as indicated by vector differences between successive demagnetization steps. Heating the specimens above 500°C commonly isolated stable end points, except for a few specimens which became magnetically unstable. After the 500°C treatment, the unstable specimens acquired large random components of viscous remanent magnetization (VRM) in the ambient field, so the results were discarded. Although AF treatment applied to companion specimens gave stable results, the isolation of a

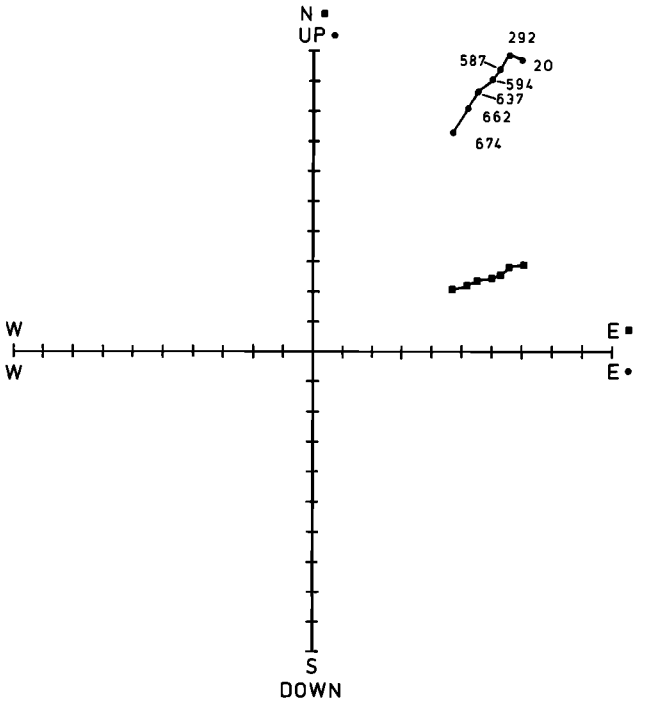


Fig. 4. Vector diagram of progressive changes in magnetic direction during thermal demagnetization of example 1 from group 1; squares, horizontal components; dots, east and vertical components; one division on axes = 4.02×10^{-2} A/m.

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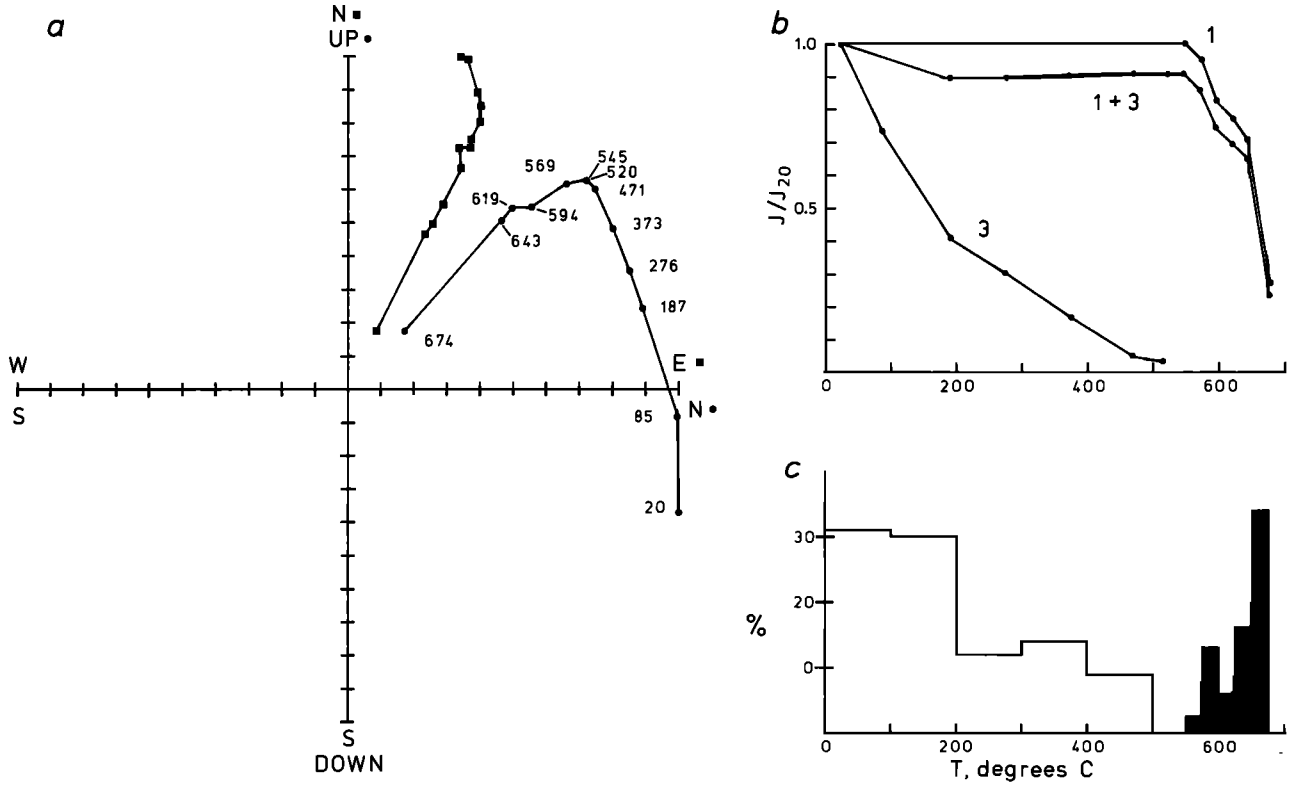


Fig. 5. (*a*) Vector diagram of progressive changes in magnetic direction during thermal demagnetization of example 2 from group 1; squares, horizontal vector components; dots, north and vertical components; one division on axes = 3.82×10^{-2} A/m. (*b*) Normalized intensities during thermal demagnetization of specimen in Figure 5*a*; 1 + 3 = total magnetization; 1 = contribution from group 1 component; 3 = group 3 overprint. (*c*) Blocking temperature distribution of group 1 magnetization (shaded) and group 3 magnetization, derived from curves in Figure 5*b*.

single end-point magnetization could not be confirmed by the vector diagrams. Nevertheless, the AF results were retained for sites 22 and 23 because they were consistent with results from the thermally stable specimens. All specimens from site 21 (metabasalt) were unstably magnetized, and the results were discarded.

Example 3, a reddish sandstone from the Huntington Formation, has thermal properties typical of the specimens that remained free of VRM during the heating experiments (Figure 7). The simultaneous removal of two magnetic components is indicated by the curved projections between J_{20} and J_{500} on the vector diagram. Separation of the two components below 500°C and calculation of corresponding T_B distributions are not possible because the distributions overlap. The magnetic direction stabilizes above 500°C; most of the stable component resides in domains having 500° $< T_B <$ 650°C. Alternating field demagnetization of the companion specimen from the same core indicates that the high-T_B component has MDF > 100 mT, near the limit of our demagnetization apparatus. Stable magnetic directions having $T_B >$ 600°C or MDF > 60 mT from the Huntington Formation make up part of group 2.

Example 4 shows the thermal properties of NRM that were typically observed in specimens from the Wild Sheep Creek Formation in the Imnaha River Canyon (Figure 8). The NRM of this specimen of metabasalt consists of two components with nonoverlapping T_B distributions: a group 2 magnetization having $T_B >$ 525° and a steep, northwesterly component having $T_B <$ 500°C. Separation of the two components by vector subtraction of J_{525} from the lower temperature steps yields the distribution of the component having lower T_B. The group 2 magnetization resides in domains having 525° $< T_B <$ 625°C; 60% of the remanence was unblocked by heating the specimen from 560° to 575°C. In contrast the low-T_B magnetization is evenly distributed in the range of blocking temperatures below 500°C.

Alternating field treatment in fields exceeding 30 mT yielded magnetic directions identical to the results obtained by heating to 560°C, as was shown by vector diagrams of the AF experiments. The stable end-point magnetization is defined by AF treatments between 30 mT and 60 mT. Less than 10% of the stable component can withstand AF demagnetization at 60 mT. Metabasalt (site 26) and red breccia (site 25) exhibited similar properties of T_B and AF stability. Accordingly, both sites were given AF treatment of 30–40 mT to isolate the stable components, which we have included in group 2.

The low T_B component of example 4 is similar to directions obtained from site 24, a basalt dike, and from four specimens within 50 cm of a thin dike that intrudes the metabasalt flow of site 26. These dikes are possibly of Tertiary age, so we have rejected results related to them. Low T_B distributions of several specimens from site 25 revealed a different component, similar to the steep northeasterly direction determined from site 27, a metabasalt flow. We interpret the component as part of group 3, described in a later section of this paper.

The remaining member of group 2 comprises approximately half of site 13 on the IPALCO road (designated site 13a). Thermal properties of the group are typified by example 5, a

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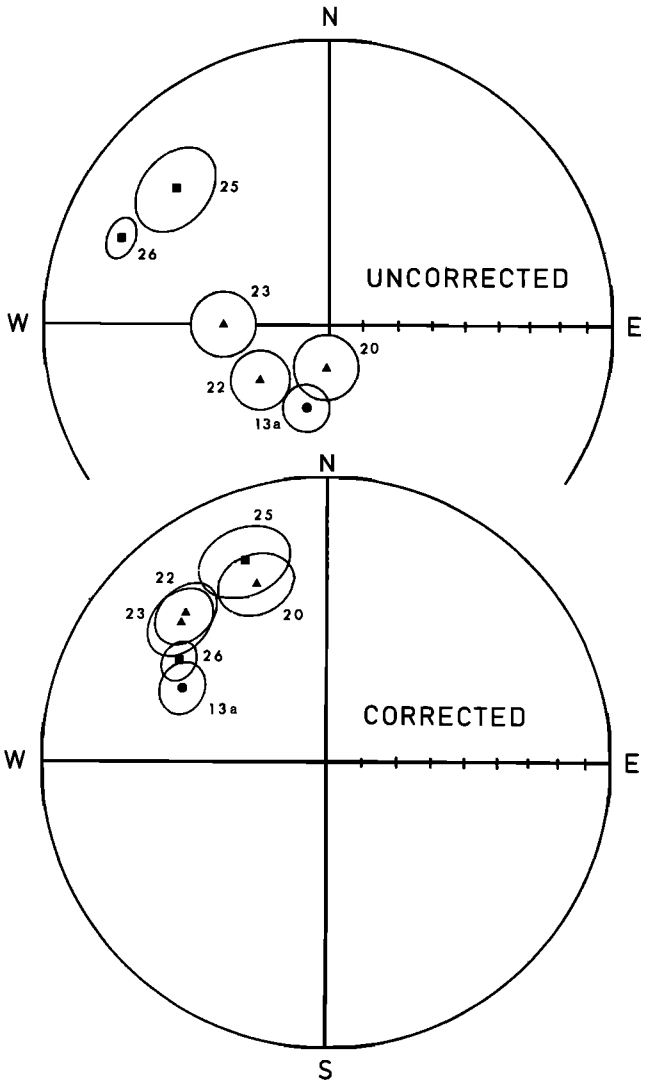


Fig. 6. Mean magnetic directions and 95% confidence limits from group 2 before and after application of the bedding corrections. All means have positive inclinations; solid circle, site in Hells Canyon; squares, sites in Imnaha River Canyon; triangles, sites in Huntington Formation of *Brooks* [1979*b*].

red volcaniclastic sandstone that has two components of magnetization (Figure 9). Distributions of the components do not overlap, so we can isolate the components by vector subtraction of J_{662} from the lower-temperature steps. The result is sharply peaked distribution centered on 670°C and a distribution below 640°C, which we attach to group 2 and 3, respectively. Alternating field treatment was ineffective in decreasing the NRM of these sandstone specimens; 90% of the initial remanence remained after application of a 100-mT field. Routine thermal cleaning of site 13, employing five steps between 575°C and 680°C, yielded 15 specimens having stable magnetizations belonging to group 2. The remaining 24 specimens reached stable end points that characterize group 3 and showed no evidence of the high T_B magnetic direction illustrated by example 5.

The five members of group 2 are within structural blocks that have enough variations in attitude to make a significant fold test (Figure 6). Correction for tilt of the bedding increases the Fisher precision parameter from 6 to 28 (Table 3). Application of the F test ($n = 4$) gives a positive result at the 95% confidence level, indicating that the decrease in directional scatter caused by the bedding correction is statistically significant. This result provides strong evidence that group 2 magnetizations were acquired before the Huntington and Wild Sheep Creek formations were deformed.

Group 3

The majority of sites give steep, northeast directions (not corrected for bedding) belonging to group 3, usually occurring as a single-component magnetization and less frequently being superimposed on group 1 and 2 components (Table 2, Figure 10). Group 3 directions are found in volcaniclastic rocks and in metabasalt flows as well. The sites are concentrated along the IPALCO road in Hells Canyon, and a few isolated sites are in the canyons of the northern Snake River and the Imnaha River.

Sedimentary rocks classified in group 3 typically consist of reddish sandstone, breccia, and tuff, all composed predominantly of volcanic detritus (sites 4, 13b–17, 19, 28, 29). Of the specimens that exhibited single-component magnetizations during thermal demagnetization, most had thermal properties similar to example 6 (Figure 11). In general, 50% of the remanence is unblocked by heating up to 500°C; the remaining 50% is unblocked between 500°C and 675°C. Alternating field demagnetization of unheated companion specimens revealed very high coercivities of NRM, as treatment in 100 mT typically reduced the magnetization less than 10%. On the basis of the pilot demagnetization experiments the remaining specimens of sedimentary rock were heated in five steps to temperatures between 575°C and 675°C to ensure that thermally stable end points had been reached. After examining the resultant vector diagram from each specimen we selected results from the heating step that best represent-

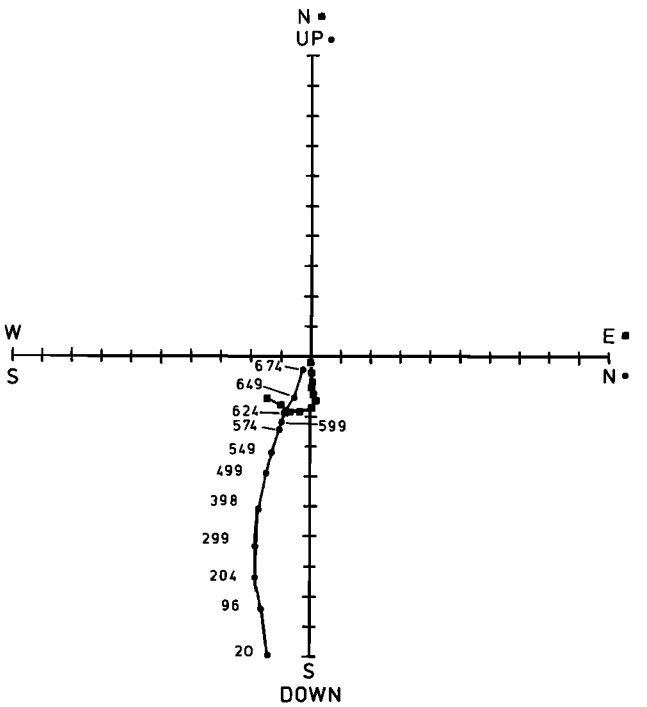


Fig. 7. Vector diagram from thermal demagnetization of example 3. Magnetization of group 2 is isolated by heating above 500°C; squares, projections of horizontal components; dots, projections of south and vertical components; one division = 1.97×10^{-3} A/m.

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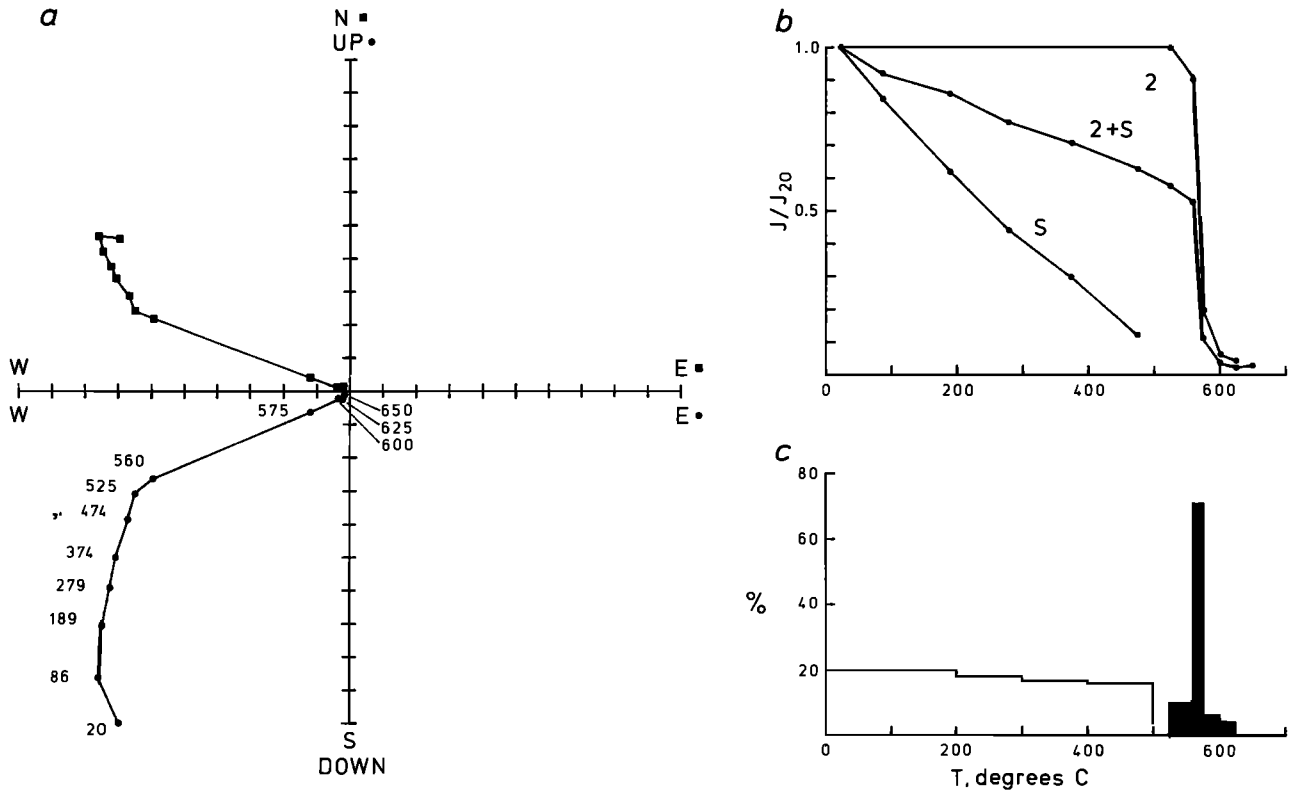


Fig. 8. (*a*) Vector diagram of example 4, showing separation of secondary magnetization from group 2 component; squares, horizontal components; dots, west and vertical components; one division = 7.58×10^{-2} A/m. (*b*) Normalized intensity plots of total magnetization (2 + *S*), secondary magnetization (*S*), and group 2 component (2). (*c*) Blocking temperature distributions of *S* and 2 (shaded) derived from curves in Figure 8*b*.

ed the end-point magnetic direction. Diagrams from all but five specimens indicated stable end points had been reached.

The volcanic rocks in group 3 (sites 1–3, 5, 6, 11, 12, 17, 27, and 29) were given treatment in AF's up to 60 mT, which typically reduced the intensity of magnetization to 1–10% of the NRM. Median destructive fields ranged from 2.5 to 20 mT for these pilot specimens. Vector diagrams and stereographic plots of vector differences between successive AF steps indicate the metabasalts are free of multicomponent magnetizations. Routine magnetic cleaning of the remaining specimens were made in alternating fields of 15 to 30 mT.

Before application of bedding corrections, the sites in group 3 yield a well-defined mean direction (inclination, 71.8°, declination, 38.6°, α_{95}, 4.6°) that is neither like the present field at the locality nor the recent axial dipole field direction (Figure 10). The mean includes only those specimens which, during thermal or AF cleaning, reached stable end points characteristic of group 3 and does not include difference vectors from specimens having group 3 magnetizations superimposed on more highly stable group 1 or 2 components. When obtained by vector subtraction, the group 3 component was poorly determined because the difference vectors were usually small in comparison to the remaining magnetization of the specimens. Therefore, such results were not included in the mean direction of group 3.

The angular dispersion of group 3 increases significantly when the results are referred to the paleohorizontal, as is shown by a decrease of the Fisher precision parameter from 60, for in situ results, to 6, for the tilt-corrected results (Figure 10, Table 3). Failure of the fold test indicates that the magnetization of group 3 was acquired after the Seven Devils arc was faulted and folded, so the magnetization cannot be the original remanence acquired during deposition. We interpret the component to be a secondary overprint that has affected both basalts and sedimentary rocks over almost the entire region sampled. In some areas the remagnetization has been complete, while elsewhere the secondary component is minor or absent; T_B distributions of the overprint are variable. Where remagnetization is complete, T_B's are rather evenly distributed between 200° and 675°C; partially remagnetized specimens usually have T_B's less than 600°C. In contrast group 1 and 2 magnetizations have sharply peaked T_B distributions between 525° and 675°C.

Magnetic Mineralogy and Rock Magnetism

The demagnetization experiments indicate that hematite is a significant carrier of NRM in many of the volcanic rocks and in all of the sedimentary rocks. The evidence for this is blocking temperature distributions between 585°C (the Curie temperature of magnetite) and 680°C (the Neel temperature of hematite), and very high coercivities of NRM during AF demagnetization. In particular, nearly all of the group 1 and 2 magnetizations after thermal cleaning, as represented in Table 2, are carried by hematite. Not all of the observed primary magnetization resides in hematite, however. In example 2 of group 1 the blocking temperature distribution is bimodal (Figure 5), and about one fourth of the high-temperature magnetization is carried by magnetite. In example 4, nearly all of the group 2 magnetization is in magnetite

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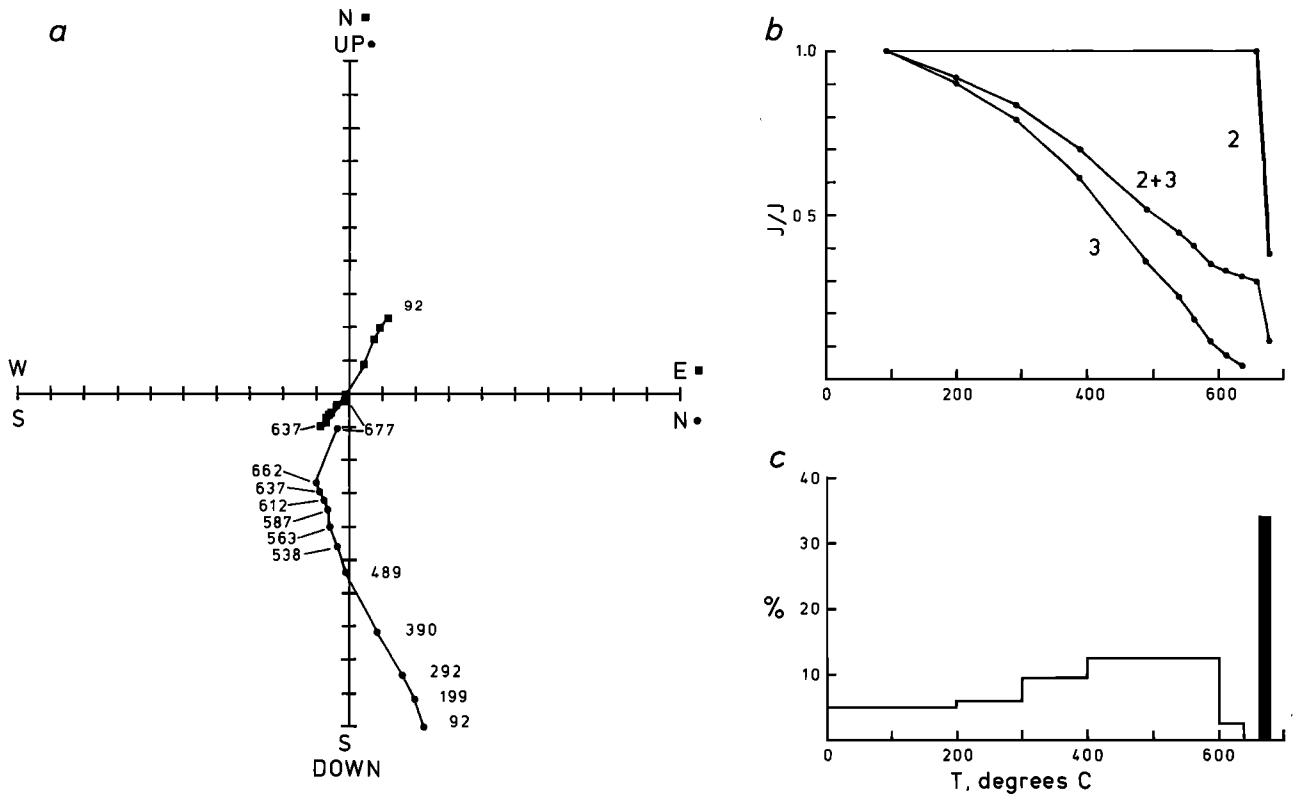


Fig. 9. (*a*) Thermal demagnetization of example 5, showing superposition of group 3 magnetization on group 2; squares, horizontal components; dots, east and vertical components; one division = 1.50×10^{-3} A/m. (*b*) Normalized intensity plots of total magnetization (2 + 3), group 2 component (2), and group 3 component (3). (*c*) Blocking temperature distributions of component 3 and component 2 (shaded).

(Figure 8). Conversely, although much of the group 3 magnetization has a moderate to low blocking temperature distribution, example 6 (Figure 11) shows that in some instances this secondary component is carried by hematite.

Several polished sections of various rock types were examined under oil immersion at magnifications up to 1050x. Abundant authigenic hematite occurs as cloudy red pigment or as microcrystals. In the volcaniclastic rocks, microcrystalline hematite that replaces hornblende and pyroxene preserves the crystal outlines of the silicate grains. Relict magnetite crystals, now almost entirely replaced by martite, are also common. In the sandstones, grains of quartz and feldspar are typically coated with microcrystalline hematite. The outlines of glass shards in one tuff specimen are beautifully preserved by microscopic hematite rinds. Although magnetite occurs only as minor remnants in the volcaniclastic specimens, it is common in the greenish metabasalts and meta-andesites. The magnetite grains are usually homogeneous, without the two-phase lattice structure commonly observed in volcanic rocks that have undergone high-temperature oxidation. Some magnetite crystals appear pitted or corroded and have faint halos of hematite pigment.

In addition to the vector analyses of demagnetization of NRM, strong-field magnetic properties were measured, using companion specimens of the examples discussed above and two more that exhibited group 3 NRM's. Figure 12*a* shows strong-field thermomagnetic curves for five specimens in which hematite is the dominant ferrimagnetic mineral. Because the specimens contained a large proportion of iron-bearing silicates in relation to the amount of hematite present, the original thermomagnetic records were dominated by the hyperbolic decrease of paramagnetism induced in these silicates. Most of the paramagnetic contribution has been subtracted from the curves shown in Figure 12*a*, assuming that the paramagnetic susceptibility follows the simple Curie law. Several curves in Figure 12*a* are concave upward between 20° and 500°C; this concavity is an artifact of the method of subtracting paramagnetism, indicating that the Curie-Weiss law would be more appropriate. Nonetheless, the procedure makes the Curie temperatures more evident, and those in Figure 12*a* are all ascribable to pure hematite within the uncertainties of measurement. Four of these five samples are red or orange volcanic sandstone, and the fifth (7) is a red tuffaceous breccia.

Strong-field thermomagnetic curves for three specimens in which magnetite is dominant are shown in Figure 12*b*. Paramagnetism was not subtracted from these curves. The main Curie temperatures range from 573° to 577°C, about 10°C below the Curie temperature of pure magnetite, which is a small but experimentally significant difference. The small peaks near 170°C for specimens 2 and 4 are due to the presence of minor pyrrhotite. These peaks are characteristic for compositions ranging from Fe_9S_{10} toward FeS, which are antiferromagnetic at room temperature [*Schwartz*, 1975]. Hence the pyrrhotite contributes nothing to the NRM.

While the thermomagnetic curves are necessary for identification of Curie temperatures and hence chemical compositions, owing to instrumental limitations they are not capable of detecting a small fraction of hematite in the presence of a large fraction of magnetite. In examples 2 and 4 (Figures 5 and 8) the group 1 and group 2 NRMs appear to be carried by both magnetite and hematite. A more sensitive experimental

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TABLE 3. Overall Mean Paleomagnetic Data From Wild Sheep Creek Formation and Huntington Formation of *Brooks* [1979*b*] and North American Paleomagnetic Poles

	N	I	D	R	k	α_{95}	θ	ϕ	A_{95}
				Group 1					
Uncorrected	5	−28.7	58.7	4.1361	5				
Corrected	5	−31.1	83.2	4.7923	19	17.9	7.6 S	350.3 E	17.0
Rotated*							25.7 N	45.3 E	17.0
				Group 2					
Uncorrected	6	61.6	271.5	5.1477	6				
Corrected	6	33.8	318.6	5.8189	28	13.0	46.6 N	130.0 E	14.0
Rotated*							1.0 N	179.1 E	14.0
				Group 3					
Uncorrected	17	71.8	38.6	16.7351	60	4.6	63.3 N	291.8 E	7.4
Uncorrected†	17	71.5	44.8	16.7351	965	4.5	60.2 N	295.3 E	7.2
Rotated*							72.6 N	200.2 E	7.2
Corrected	17	54.5	0.1	14.4855	6	15.4	82.6 N	70.9 E	13.8
				Paleomagnetic Poles					
Plutons of Late Jurassic and Early Cretaceous age [*Wilson & Cox*, 1980]							68.5 N	320.9 E	12.9
Plutons of Late Jurassic and Early Cretaceous age rotated*							65.0 N	165.2 E	12.9
Late Jurassic and Early Cretaceous: North America							65.6 N	175.0 E	6.9
Late Triassic: North America							65.3 N	94.2 E	5.9

N, number of sites; I, D, mean inclination and declination; R, resultant vector; k, α_{95}, Fisher precision parameter and 95% confidence limit for directions; θ, ϕ, A_{95}, latitude, longitude, and 95% confidence limit for VGP's.

*Rotation is 66.4° counterclockwise around mean site location.

†Two-level analysis [*Watson and Irving*, 1957] of directions and poles not corrected for tilt of bedding (number of specimens, 182).

method for determining a minor amount of one oxide in the presence of the other is the mode of acquisition of isothermal remanent magnetization (IRM) in progressively higher applied direct fields; IRM curves for the specimens in which hematite is abundant are shown in Figure 13*a*. In all these examples, saturation of IRM has not occurred at applied fields up to 700 mT, which is diagnostic for hematite in the light of the thermomagnetic curves. Ferrimagnetic pyrrhotite (Fe_7S_8) would also be expected to show such high coercivity [*Schwarz*, 1975], but there is no evidence for it in the thermomagnetic curves nor would it be expected to be present in such highly oxidized rocks. In three of the examples shown in Figure 13*a* (3, 5, and 7) the IRM curves show slight upward convexities between 0 and 150 mT. These may be interpreted as indicating traces of magnetite.

Figure 13*b* shows IRM curves for the three examples for which the thermomagnetic curves indicated low-Ti titanomagnetite (loosely designated 'magnetite' herein). In example 8, which is a metabasalt, the IRM is completely saturated at 400 mT, meaning that hematite is not present in any significant amount. Conversely, the IRM in example 2 is not saturated at 700 mT, which signifies the presence of minor hematite, and example 4 may also contain a trace of hematite.

The results of all the rock-magnetic experiments are summarized in Table 4, in which are also included the values of the median destructive field (MDF) of NRM. Where the MDF is shown as >100 mT in Table 4, this means that in fact AF demagnetization to this field strength (120 mT in one instance) did not reduce the NRM by a measurable amount. Such a high coercivity of NRM is typical of hematite. For the most part the data in Table 4 are self consistent, that is, the thermomagnetic and IRM results match the inferences drawn from thermal and AF demagnetization of NRM. The only apparent inconsistency is example 2, in which the Curie temperature and IRM data indicate mainly magnetite, but the MDF and the blocking temperatures indicate that most of the group 1 NRM is carried by hematite. In other words, in example 2 the trace of hematite seen in the IRM curve (Figure 13*b*) is responsible for most of the total NRM (group 1 and the unspecified secondary component). Note also that in two of the group 3 examples, 6 and 7 in Table 4, the NRM resides in hematite.

Origin of Remanent Magnetization

The Seven Devils and Huntington arcs record two major episodes during which magnetization was acquired. The older episode, defined by groups 1 and 2, occurred before the region was faulted and folded in the Late Jurassic [*Brooks et al.*, 1976; *Dickinson and Thayer*, 1978; *Brooks and Vallier*, 1978]. This conclusion is substantiated by positive fold tests. We interpret groups 1 and 2 as magnetizations that were acquired during normal and reversed polarity epochs of similar age. A comparison of the two groups shows that the inclinations have equal magnitudes and opposite signs; however, the groups deviate from antipolarity by about 40° in declination (Figures 3 and 6). Within each group, declinations vary more than inclinations (I) when the $1/\cos(I)$ dependence of the declination variation is taken into account. The deviation from antipolarity can be explained by local rotations of three structural blocks represented by the Huntington section, the Imnaha River sites, and the Hells Canyon reservoir sites. The observed antipolarity of the group 1 mean direction to the result from site 13b, all within the Hells Canyon block, supports this explanation. Also, when folding is about nonhorizontal axes, or when oblique

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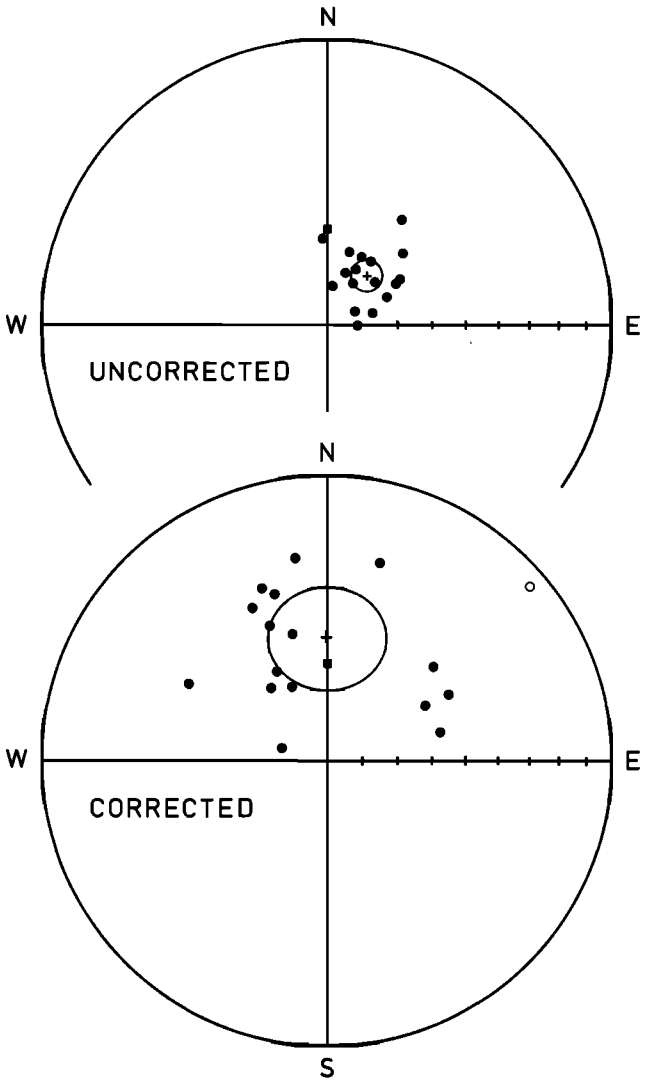


Fig. 10. Mean magnetic directions (circles) from sites in group 3, before and after bedding corrections. Solid symbols have positive inclinations; plus, overall site mean; square, present axial dipole field direction.

slip occurs on faults, the simple bedding correction will introduce errors in declination but not in inclination.

Group 1 and 2 magnetizations were probably acquired during the Late Triassic, during or soon after the volcanic and volcaniclastic rocks were deposited. The magnetization resides in authigenic hematite and to a lesser degree in magnetite. The hematite is probably a chemical product of intense oxidation that accompanied the influx of hot volcanic material into a near-surface marine environment. Many samples are from reddened sediments that have been baked by overlying basalt flows. The resulting authigenic hematite would acquire stable remanence as TRM caused by heating or as chemical remanent magnetization (CRM) caused by growth of single-domain hematite crystals [*Stacey and Banerjee*, 1974], while detrital magnetite grains would acquire TRM.

Group 3, which failed the fold test, represents the younger magnetization episode. Carried by magnetite and authigenic hematite, the component probably developed during the intrusion of Late Jurassic and Early Cretaceous plutons such as the Wallowa batholith, the Cornucopia stock, and the Deep Creek stock. Potassium-argon ages of these plutons range from 160 m.y. to 126 m.y., and Rb-Sr whole rock isochron ages are 134 m.y. and 133 m.y., respectively [*Armstrong et al.*, 1977]. This interval is marked by widespread plutonism, uplift, and deformation throughout the Blue Mountains province [*Dickinson and Thayer*, 1978]. We ascribe the magnetic overprint to this plutonic episode because the corresponding paleomagnetic pole is nearly identical to poles obtained from the Wallowa batholith and Cornucopia stock, which intrude the Seven Devils arc [*Wilson and Cox*, 1980] near our sampling localities. The acquisition of the overprint is confined to a single normal polarity epoch, which could be as long as 3 million years, according to *Larson and Hilde*'s [1975] time scale for the period.

Hydrothermal alteration caused by the plutons is probably responsible for the magnetic overprint that we observed. During this episode of plutonism, the Wild Sheep Creek Formation was buried beneath at least 2 km of sedimentary rocks [*Vallier*, 1977]. As magma intruded this water-laden sedimentary blanket, hydrothermal circulation would begin in the more permeable strata surrounding each pluton [*Norton and Knight*, 1977]. Isotopic measurements taken in rocks surrounding a number of plutons in western Oregon show significant decreases in $^{18}O/^{16}O$ up to three pluton diameters from the intrusive contacts [*Taylor*, 1971]. The depletion of ^{18}O in the country rock is attributed to oxygen equilibrium between the rock and vast volumes of meteoric water circulated by the hydrothermal system around the pluton. During this process, primary igneous pyroxenes, olivine, and hornblende are commonly altered to chlorite, epidote, and iron oxides. Such hydrothermal systems were probably at work in the Seven Devils Group, because we have observed chlorite, epidote, and hematite pseudomorphs of primary ferromagnesian minerals. In addition to oxidizing iron-bearing minerals in situ, the circulating water carries iron in solution until conditions of E_h and pH are favorable for the precipitation of hematite and goethite. Hematite rinds on quartz grains and linings of pores may form in this manner. As microcrystalline hematite grows beyond single-domain grain sizes, an extremely stable CRM will develop in the direction of the ambient field. For the Seven Devils region it is reasonable to assume that cessation of the remagnetization process and the shut-off of hydrothermal circulation would coincide with the commencement of radiogenic argon retention about 125–135 m.y. ago. Secondary magnetite commonly forms along with chlorite and epidote as byproducts of greenschist metamorphism in volcanic rocks. In areas where the magnetic overprint is carried by magnetite, heat generated by the plutons may have created new TRM in either original or secondary magnetite grains.

Our results show that the degree of remagnetization was highly variable throughout the region. For example, the older component and the overprint, both carried by hematite, were found in specimens at few meters apart at site 13. In contrast we found zones in Hells Canyon where the remagnetization was complete over distances of several hundred meters. If the magnetic overprint is largely CRM that developed from the oxidation of iron-bearing minerals and by the redistribution of iron during hydrothermal circulation, then the spottiness of the remagnetization process can be explained by variable permeability of the rock. The permeability is controlled by pore space and, probably to a

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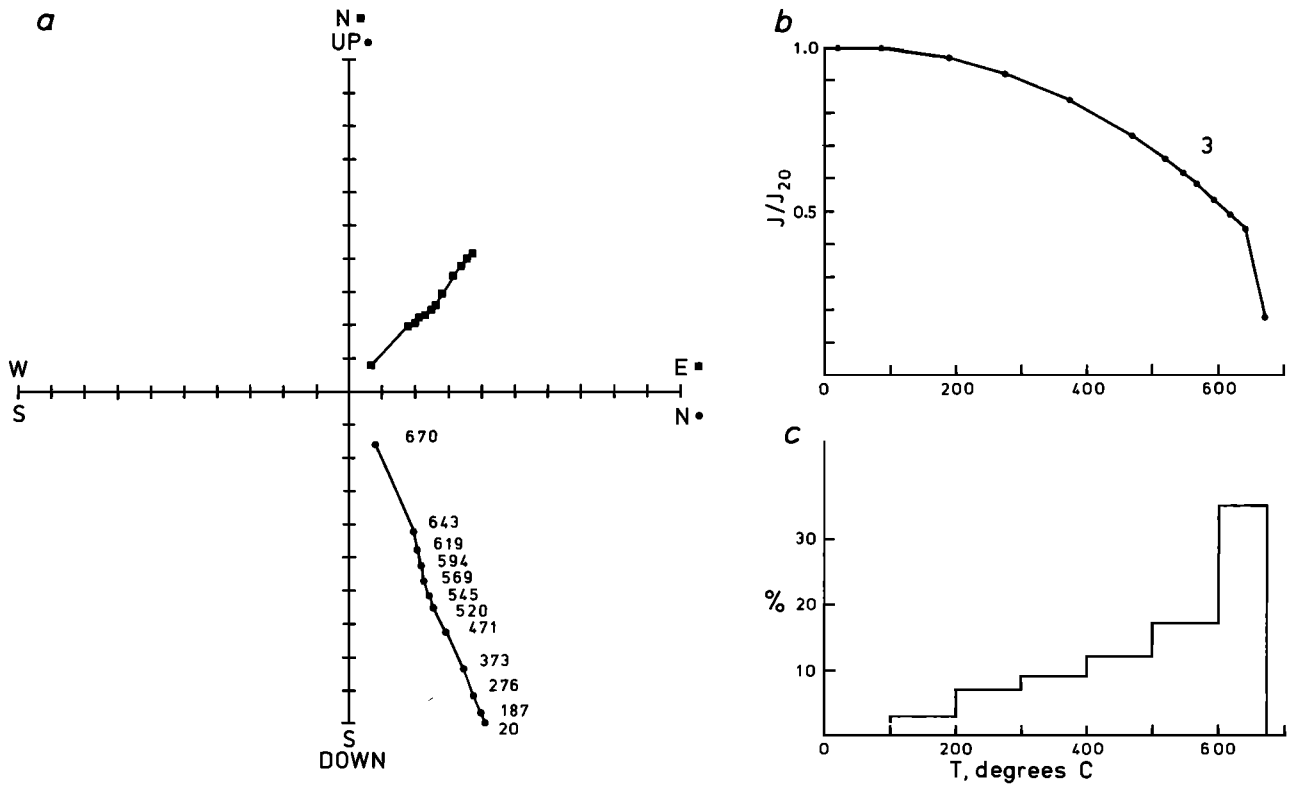


Fig. 11. Example 6, showing thermal properties of completely remagnetized samples in group 3. (*a*) Vector diagram, one division = 2.76×10^{-3} A/m. (*b*) Normalized intensity plot. (*c*) Blocking temperature spectrum derived from Figure 11*b*.

larger degree, by fractures. Areas isolated form the hydrothermal fluid by low permeability would preserve the older magnetization; highly permeable areas would not. Another critical factor affecting the precipitation of hematite is the oxidation potential of the iron-bearing fluid. Iron would be removed from some areas to be recrystallized in hematite elsewhere, depending on the *pH and* E_h of the fluid.

Tectonic Implications

We have calculated a mean paleolatitude from groups 1 and 2, assuming that the results represent the axial dipole field at the time of deposition and magnetization. Positive fold tests, the attainment of stable end-point magnetizations, and consistent normal and reversed inclinations justify this assumption. We did not use Fisher statistics to derive the mean paleolatitude and confidence limits from the magnetization vectors because local rotations and the bedding corrections probably introduced artificially large declination variations. Instead, we converted the mean inclination from each site to a corresponding paleolatitude and calculated the mean paleolatitude and 95% confidence limits using *Kono*'s [1980] statistical method. The resulting mean distance from the paleoequator is 18° ± 4°, but the paleolatitude may lie in either hemisphere because the true polarity of group 1 or 2 is unknown. The ambiguity cannot be resolved from the declinations because large-scale rotations are highly possible in the Cordillera.

The inferred paleolatitude, whether it lies in the northern or southern hemisphere, is anomalously low compared to the paleoposition of Hells Canyon during Late Triassic and Early Cretaceous time, on the basis of North American apparent polar wander [*Irving*, 1979]. We minimize the discrepancy in paleolatitude by assuming that the magnetization is contemporaneous with deposition of the rocks during the Late Triassic. Therefore, we selected the Late Triassic paleomagnetic pole (65.3°N, 94.2°E, α_{95} = 5.9°) from central North America as a reference for determining northward drift of the Seven Devils and Huntington arcs (Table 3). The reference pole is the mean of I 8.44, I 8.62, I 8.63, I 8.111, I 8.213, and I 8.237 given in the paleomagnetic catalogs of *Irving et al.* [1976] and *Hicken et al.* [1972]. Given an expected paleolatitude of 23° ± 6° for Hells Canyon, the inferred northward drift is 6°, if the arcs originated in the northern hemisphere, or 42° if they originated in the southern hemisphere. The confidence limit (7°) of the northward displacement is obtained by taking the square root of the sum of the squares of the 95% confidence limits of the expected and observed magnetic latitudes.

A comparison of paleomagnetic results from the Seven Devils and Huntington arcs with data from other Triassic sequences in Wrangellia suggests that all these regions once shared a common paleolatitude. Triassic paleolatitudes have been determined from the Karmutsen Formation (13° to 18° N or S) on Vancouver Island [*Yole and Irving*, 1980; *Schwarz et al.*, 1980], the Nikolai Greenstone (15°N or S ± 14°) near McCarthy, Alaska [*Hillhouse*, 1977], and Nikolai-equivalent basalt (18° N or S ± 8°) in the Clearwater Mountains 250 km northwest of McCarthy [*Hillhouse and Gromme*, 1980]. The northern and southern extremes of Wrangellia as presently distributed apparently originated in a limited region located approximately 18° from the Late Triassic paleoequator. For comparison we have calculated the expected paleolatitudes of the various parts of Wrangellia, using the Late Triassic reference pole for stable North

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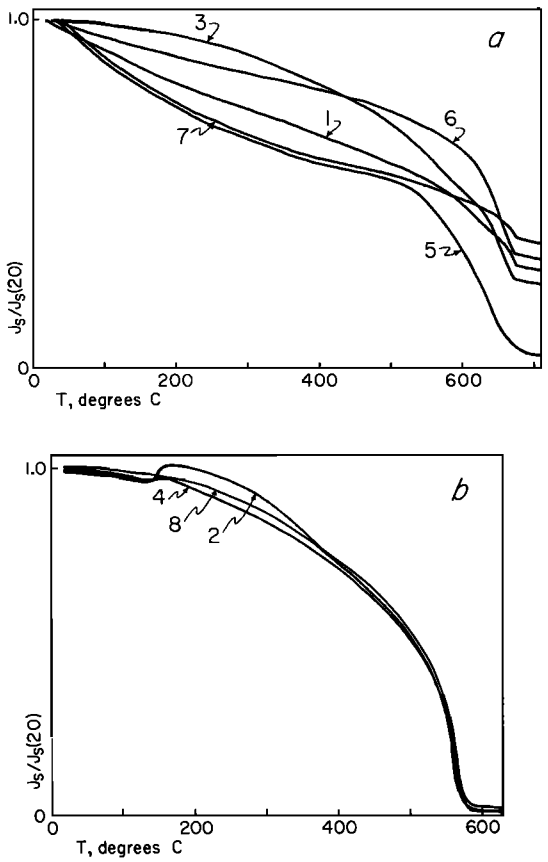


Fig. 12. Strong-field thermomagnetic curves. (*a*) Examples containing hematite. Heatings done in air; magnetizing fields range from 400 to 700 mT. (*b*) Examples containing low-Ti titanomagnetite; heatings done in nitrogen (examples 2 and 8) or argon (example 4); magnetizing fields range from 200 to 300 mT. Example numbers are keyed to Table 4.

America. The expected latitudes range from 23°N for the Seven Devils region to 46°N for Clearwater Mountains, Alaska, all having 95% confidence limits of ±6°. The differences between expected and observed paleolatitudes indicate that parts of Wrangellia have subsequently moved large distances north relative to the craton. If Wrangellia originated at 18°N, then the Alaskan fragments have drifted north approximately 28°; inferred northward drift of the southern fragments is 6°–10°, but is less than the 95% confidence limits of the individual paleomagnetic studies.

An additional source of error, which may account for up to 5° of the discrepancy between predicted and observed paleolatitudes, is posed by the age differences between the reference pole and the volcanic rocks studied. The Karnian volcanic rocks of Wrangellia may be up to 10 m.y. older than the Newark Group of the eastern United States, data which strongly influence the position of the Late Triassic reference pole. Apparent polar wander in the Triassic has the right sense and magnitude to account for the low paleolatitude observed in the Karmutsen Formation and the Seven Devils arc, provided that the terranes originated in the northern hemisphere. However, apparent polar wander cannot account for the discrepant paleolatitudes if the rocks originated in the southern hemisphere, nor can it explain the results from the Alaskan fragments of Wrangellia, regardless of the hemisphere in which the rocks originated.

Yole and Irving [1980] favored the option that the Karmutsen Formation originated in the southern hemisphere, because the northern option required a rotation of 180° about a vertical axis to account for the Karmutsen's observed northerly paleomagnetic declination. If their hypothesis is correct, then northward drift of Wrangellia ranges from 42° to 64° for the southern and northern ends, respectively. These great circle angles are minimum distances, which do not encompass possible, but undetectable, east-west displacements.

Paleomagnetic poles determined from the 150–130 m.y. old plutons and the group 3 overprint provide clues to the geometry and timing of final accretion of the Seven Devils arc to the craton. Accretion probably postdates the Upper Triassic and Lower Jurassic Hurwal Formation because the continuous record of depostion in the arc shows no evidence of widespread deformation during Late Triassic and Early Jurassic time [*Davis et al.*, 1978]. An upper limit on the age of accretion is set by middle Cretaceous sedimentary rocks, which indicate that the older rocks of the Blue Mountains had been uplifted, shedding granitic and volcanic debris into shallow basins of the continental margin [*Dickinson and Thayer*, 1978]. From the paleolatitude of the Wallowa batholith and related stocks, *Wilson and Cox* [1980] concluded that the Seven Devils volcanic arc had arrived at its present latitude, relative to the craton, by about the time previously set for the Jurassic-Cretaceous boundary (138 m.y.). We transformed their published magnetic directions to virtual geomagnetic poles (VGP) and calculated a mean pole for comparison with our paleomagnetic pole from group 3 and the coeval reference pole for North America (Figure 14, Table 3). The reference pole includes the following Late Jurassic and Early Cretaceous paleomagnetic poles: 9-176, 10-12, 10-72, 10-73, 10-35, 10-17, and 10-256 from the Ottawa list [*Irving et al.*, 1976], and it differs slightly from *Irving*'s [1979] reference poles (130 and 140 m.y.) used by Wilson and Cox. Our reference pole represents the time interval from 140 to 100 m.y. Although Early Cretaceous poles are well represented, the pole is poorly known for the Late Jurassic, when apparent polar wander was quite rapid. For the plutons and the group 3 overprint, differences between the expected and observed magnetic latitudes are well within the large 95% confidence limits of the corresponding poles, suggesting that the northward drift of the Seven Devils arc was complete by the Early Cretaceous.

Wilson and Cox [1980] explained the difference between the Jurassic-Cretaceous reference pole for North America and the pole they obtained from the Wallowa batholith and related rocks as resulting from a postintrusion clockwise rotation of the entire sampling area around a vertical axis. The rotation they calculated, expressed as a difference between expected and observed paleomagnetic declinations, was 60° ± 29° clockwise. We have repeated the calculation, using the revised Jurassic-Cretaceous pole and the paleomagnetic pole obtained from the data of *Wilson and Cox* [1980] described above. The calculation may be performed in 'direction space,' as was done by Wilson and Cox, or in 'VGP space,' as discussed and illustrated by *Yole and Irving* [1980] (their model B). The two methods will obviously give the same rotation, but in terms of Fisher statistics, using 'VGP space' will give more accurate uncertainties if VGP's are more nearly circularly distributed than are paleomagnetic directions. The rotation we obtained from the VGP's is 66.4° ± 21.2°. Making the same calculation by using the revised reference pole and our group 3 pole (obtained by

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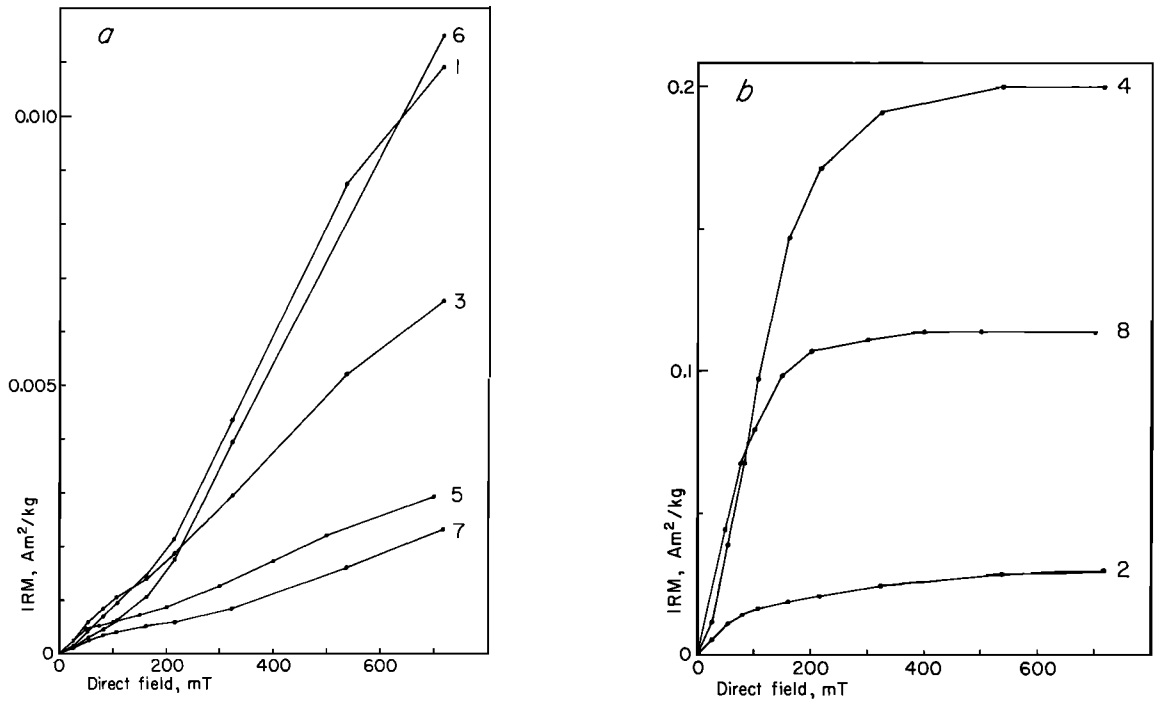


Fig. 13. Acquisition of isothermal remanent magnetization. Specimens were magnetized in progressively higher direct fields and measured after removal from the field. (*a*) Specimens containing mostly hematite. (*b*) Specimens containing mostly low-Ti titanomagnetite.

two-level analysis, Table 3) yielded a clockwise rotation of 80.6° ± 16.8°, which is not significantly different from the rotation inferred from the plutons. We believe that the rotation obtained from the plutons is a more reliable estimate because the plutons were dated directly and the distribution of paleomagnetic sites covers a larger area than does our study of the Seven Devils Group. The effect of applying the inverse of this rotation (66.4°) is shown in Figure 14. The pole for the Wallowa batholith and related rocks and the group 3 pole from the Seven Devils Group become statistically indistinguishable from the Jurassic-Cretaceous reference pole.

Alternatively, the anomalous paleomagnetic poles may be explained by westward tilt of the entire sampling area, amounting to about 30° about a NNE horizontal axis. The tilt hypothesis cannot be ruled out completely, because the true paleohorizontal plane of the plutons and of the country rock at the time of remagnetization is not known. *Wilson and Cox* [1980] preferred the rotation hypothesis, because their sample localities span 150 km of the Blue Mountains province, so that large-scale tilt would place the province's western end an implausible 75 km below the eastern end. Although the tilt hypothesis is feasible if the province has been broken into uniformly tilted blocks, the uniform tilt of many crustal blocks would be a remarkable coincidence.

Large-scale tectonic rotation of the Seven Devils arc about a vertical axis makes sense in the context of regional structural trends. The great bend of eugeosynclinal belts around the Columbia embayment, which interrupts the northwest trends of the Klamath Mountains and central British Columbia, is a well-known anomaly [*Carey*, 1958; *Hamilton*, 1978]. The sense and magnitude of the rotation implied by the paleomagnetic data are consistent with the anomalous trend of the Blue Mountains province. Timing of the rotation is constrained by paleomagnetic results from the western end of the province in the Clarno Formation of Eocene age, which has rotated clockwise approximately 15° since 50 m.y. ago (M. E. Beck, Jr. and S. Gromme,

TABLE 4. Summary of Magnetic Properties of Selected Specimens

Example	Group	Site	T_B Range Group 1	T_B Range Group 2	T_B Range Group 3	S	T_C	MDF, mT	IRM Type
1	1	8	587°–674°	—	—	20°–490°	680° ± 5°	>100 mT	hematite
2	1	7	545°–675°	—	20°–520°	—	573°	>100 mT	magnetite (trace hematite)
3	2	20	—	549°–674°	—	20°–499°	670° ± 5°	11 mT and >100 mT	hematite (trace magnetite)
4	2	26	—	525°–650°	—	20°–525°	576°	16 mT and >100 mT	magnetite (trace hematite?)
5	2	13a	—	662°–677°	20°–637°	—	670° ± 10°	20 mT and >100 mT	hematite (trace magnetite)
6	3	13b	—	—	20°–670°	—	672°	n.d.	hematite
7	3	16	—	—	97°–676°	—	675° ± 10°	>120 mT	hematite (trace magnetite)
8	3	6		n.d.			577°	10 mT	magnetite

T_B, range is of blocking temperatures derived from vector analysis of thermal demagnetization of NRM, in degrees C; S refers to secondary magnetization that is not parallel to group 3 direction; T_C is Curie temperature obtained from strong-field thermomagnetic curves, in degrees C; MDF is median destructive field obtained from AF demagnetization of NRM; higher of paired values refers to group 2 direction; IRM type is the magnetic mineralogy inferred from the shapes of IRM acquisition curves.

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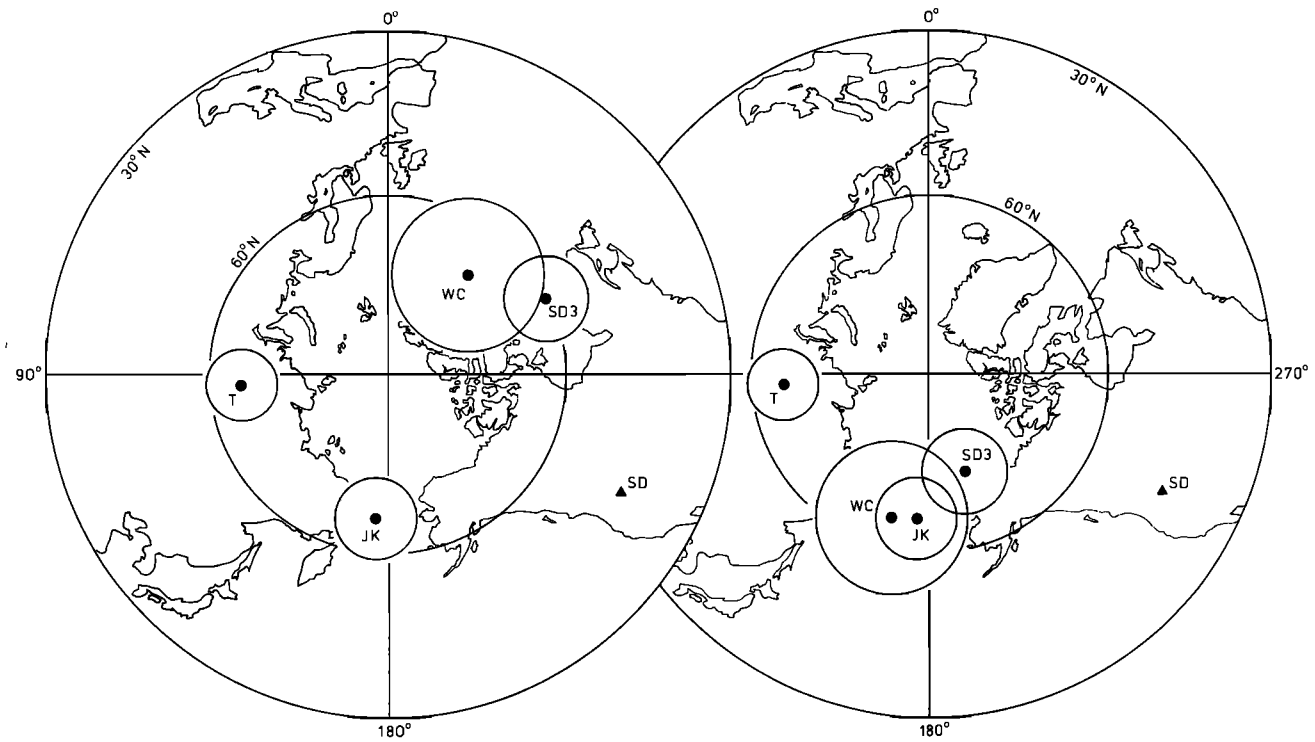


Fig. 14. Equal-area projections of northern hemisphere showing paleomagnetic poles (solid circles) with 95% confidence circles and sampling site for Seven Devils Group and Huntington Formation of *Brooks* [1979*b*] (triangle); SD 3, pole for group 3; WC, pole calculated for Jurassic and Cretaceous intrusive rocks from data of *Wilson and Cox* [1980]; JK, Jurassic-Cretaceous reference pole; T, Late Triassic reference poles. Left diagram shows observed poles. On right diagram, poles SD 3 and WC have been rotated counterclockwise around site SD so that the great circle connecting WC and SD also passes through JK.

unpublished data, 1982). Therefore, at least 45° of the apparent 66° rotation occurred between 130 and 50 m.y. ago.

An interesting comparison can be made between the paleomagnetic poles obtained from the older magnetization (groups 1 and 2) of the Seven Devils arc and the poles obtained from the coeval Karmutsen Formation of Vancouver Island, when the effect of the inferred late Mesozoic rotation is removed (Figure 15). Applying the inverse rotation to the Seven Devils poles produces excellent agreement between the Karmutsen pole of *Yole and Irving* [1980], the 'north up' Karmutsen pole of *Schwarz et al.* [1980], and 'group 1' Seven Devils pole. The antipole of the 'group 2' pole is moved closer to the main body of Karmutsen poles but ends up a considerable distance west of the cluster. This discrepancy, however, is not significantly greater than the dispersion of poles within the Karmutsen, as shown by the 'northwest-up' pole of *Schwarz et al.* [1980]. The source of the dispersion is probably block faulting within the bounds of each paleomagnetic study. We can infer from Figure 15 that the late Mesozoic rotation of the Blue Mountains is real and that the characteristic magnetizations of the Triassic Seven Devils arc and the Karmutsen Formation are very similar. The similarity supports our contention that these two volcanic provinces were once linked together as part of Wrangellia. The preponderance of northerly declinations and upward inclinations (in a Triassic frame of reference) supports the arguments for the origin of Wrangellia in the southern hemisphere. If this is so, then all of the Karmutsen data and our group 1 data represent normal geomagnetic polarity, and our group 2 data represent reversed polarity.

CONCLUSIONS

Late Triassic volcanogenic rocks of the Seven Devils Group and the Huntington Formation of *Brooks* [1979*b*] record two magnetization episodes: The older magnetization, which is carried by magnetite and authigenic hematite, predates Late Jurassic deformation of the region. We believe the remanence was acquired in the Late Triassic, during normal and reversed polarity intervals. The younger magnetization, which fails the fold test, is a widespread magnetic overprint caused by the hydrothermal circulation set up by the Wallowa batholith and related plutons of Late Jurassic and Early Cretaceous age. The overprint, mainly carried by authigenic hematite, may have formed as CRM by the oxidation of ferromagnesian mineral grains and the precipitation of microcrystalline hematite in pores. The normally polarized overprint is unlike the recent axial dipole field direction and is nearly identical to directions obtained from the plutons and their contact aureoles.

Using partial thermal demagnetization, we were able to separate the older magnetization from the overprint in 30% of the collection. Blocking temperature distributions of the older component ranged from 525°C to 680°C. In the remaining part of the collection, completely remagnetized specimens had variable blocking temperature distributions that ranged from room temperature up to 680°C. After thermal cleaning, specimens in the older magnetization group showed no evidence of the overprint; the cleaned directions are not streaked toward the overprint, and vector diagrams indicate that stable end points were reached during demag-

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netization. In addition, normal and reversed inclinations have consistent magnitudes, indicating the steeply inclined secondary component has been removed.

From the inclinations of the older magnetic component, we infer that the Seven Devils volcanic arc and the Huntington arc originated within 18° ± 4° of the Late Triassic equator as part of Wrangellia. Fragments of Wrangellia are now distributed along the continental margin from Oregon to south-central Alaska [*Jones et al.*, 1977]. Whether Wrangellia originated near 18°N or 18°S remains an unsolved problem. However, paleodeclinations from the Karmutsen Formation on Vancouver Island support the hypothesis that Wrangellia originated in the southern hemisphere [*Yole and Irving*, 1980], and our results also tend to confirm this.

The Seven Devils volcanic arc is apparently a displaced terrane that was carried to its present location by relative plate motions rather than being built by volcanic activity directly on the margin of the craton. By Late Jurassic time, the Seven Devils arc had amalgamated with the oceanic crust melange and the forearc basin terrane that is described by *Brooks et al.* [1976], *Brooks and Vallier* [1978], and *Dickinson and Thayer* [1978]. This amalgamated fragment of crust was near its present latitude in relation to cratonic North America. At that time, Triassic volcanogenic rocks now found on Vancouver Island and in Hells Canyon may have formed a continuous, north trending belt. Plutons and remagnetized country rock in and near the Seven Devils volcanic arc yield VGP's that indicate the arc has rotated clockwise approximately 66° since Early Cretaceous time. The Clarno Formation shows approximately 15° of clockwise rotation, so most of the arc's rotational movement occurred before 50 m.y. ago. Figure 15 shows that if we assume a vertical rotation axis and undo the rotation of the Seven Devils arc, we obtain a better alignment with the Late Triassic declinations determined on Vancouver Island [*Yole and Irving*, 1980; *Schwartz and others* 1980]. This observation supports our speculation of a formerly continuous Seven Devils–Karmutsen belt.

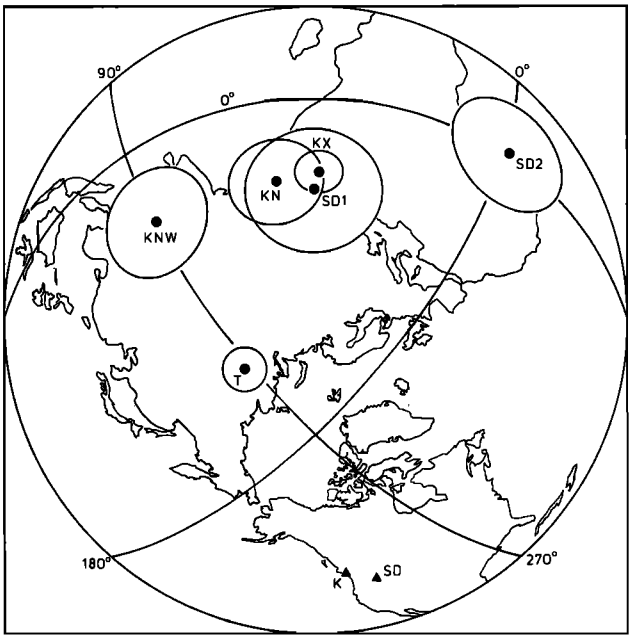


Fig. 15. Oblique equal-area projection showing paleomagnetic poles with 95% confidence circles for Seven Devils Group and Huntington Formation of *Brooks* [1979*b*] and Karmutsen Formation (circles) and respective sampling areas (triangles); SD 1, pole for group 1 NRM's; SD 2, pole for group 2 NRM's (antipole of that given in Table 3); KX, 'X' NRM of Karmutsen Formation [*Yole and Irving*, 1980]; KN, 'north up' NRM of Karmutsen Formation [*Schwarz et al.*, 1980]; KNW, 'northwest up' NRM of Karmutsen Formation [*Schwarz et al.*, 1980]; SD, sampling area of the Seven Devils Group and Huntington Formation of *Brooks* [1979*b*]; K, sampling area of the Karmutsen Formation. Poles SD 1 and SD 2 have been rotated counterclockwise around site SD as illustrated in Figure 14.

Several tectonic models have been proposed as explanations of the apparent clockwise rotation of the Blue Mountains province [*Carey*, 1958; *Hamilton*, 1978; *Wilson and Cox*, 1980]. We favor a scenario in which the combined Seven Devils–Karmutsen arc, a fragment of Wrangellia, arrived at the mouth of what is now the Columbia embayment in the Early Cretaceous. The final stage of our model involves Cretaceous breakup of the exotic arc into two pieces; the northern half remained lodged against the Pacific margin of British Columbia, while the fragment that became the Blue Mountains rotated into the Columbia embayment. Closure of the embayment requires a left-lateral fault at its northern boundary and a subduction zone on the southeast margin of the embayment. During the Cretaceous, other fragments of Wrangellia accreted to the continental margin in western British Columbia, in southeastern Alaska, and in south-central Alaska south of the Denali fault.

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